Files
main
Go to file
README.md

rohinig-faculty Update README.md ebc5dd7 · now History

EC1801-DLC-EXP1

EXPERIMENT 1

MINIMIZATION AND IMPLEMENTATION OF BOOLEAN FUNCTION

Aim

To minimize and implement a Boolean function using logic gates and verify the functionality using Synopsys VCS and DVE.

Software Required

• Synopsys VCS • Synopsys DVE • Linux Terminal • Verilog HDL

Boolean Function

Consider the Boolean function: F(A,B)=Σm(0,1,3) The minimized Boolean expression is: F=A'+B The function is implemented using Verilog HDL.

Files Used

File Description fun.v Verilog design file tb.v Verilog testbench fun.vcd VCD waveform dump generated during simulation

Verilog Program

```
//gedit fun.v
```

module fun (input wire A, input wire B, output wire F); assign F = (~A) | B ; // F = A' + B endmodule

TestBench Program

// gedit tb.v module tb; reg A, B; wire F; // Instantiate the design under test (DUT) fun uut (.A(A), .B(B), .F(F));

```
initial begin
    // ---- VCD dump setup ----
    $dumpfile("fun.vcd");   // name of the VCD file to be generated
    $dumpvars(0, tb);   // dump all signals in this testbench hierarchy

    // ---- Apply all 4 input combinations ----
    $monitor("Time=%0t A=%b B=%b | F=%b", $time, A, B, F);
```

basd A = 0; B = 0; #10; A = 0; B = 1; #10; A = 1; B = 0; #10; A = 1; B = 1; #10;

```
    #10 $finish;
end
```

endmodule

Truth Table

A	B	A'	F = A'+B
0	0	1	**1**
0	1	1	**1**
1	0	0	**0**
1	1	0	**1**

Simulation Procedure

STEP 1 – Open Terminal Open a terminal in the experiment folder. bash

STEP 2 – Load Synopsys Environment source /synopsys/start.sh

STEP 3 – Compile Using VCS vcs exp1_boolean_min.v exp1_boolean_min_tb.v -full64

If compilation is successful, VCS generates the simulation executable: simv

STEP 4 – Run Simulation ./simv The terminal displays the input combinations and corresponding output F. A VCD waveform file is also generated: exp1_boolean_min.vcd

STEP 5 – Open DVE dve -full64 Other option dve -full64 & A DVE environment will open.

DVE Waveform Verification

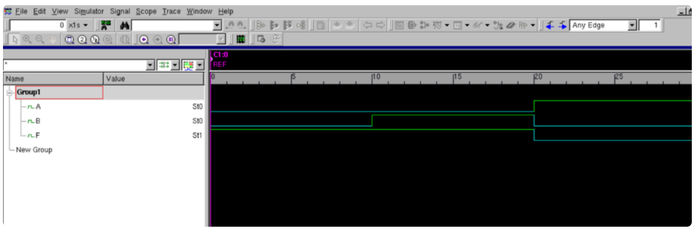
In DVE:

1. Open the testbench hierarchy.
2. Locate the signals: o A o B o C o D o F
3. Add the signals to the waveform window.
4. Run/inspect the waveform.
5. Verify that F = 1 whenever B and D are equal.
6. Verify that F = 0 whenever B and D are different. The waveform should agree with the truth table.

Expected Result

The Boolean function F(A,B)=Σm(0,1,3) minimized to F=A'+B and successfully implemented using Verilog HDL.

OUTPUT



The design was compiled and simulated using Synopsys VCS, and the functionality was verified using DVE waveform analysis. Result Thus, the Boolean function was minimized, implemented using Verilog HDL, successfully simulated using Synopsys VCS, and verified using Synopsys DVE.